Proof that $\sqrt{7}$ is Irrational

Problem

Prove that $\sqrt{7}$ is irrational.

Solution

1 Irrationality of $\sqrt{7}$sqrt(7)

The number $\sqrt{7}$ is irrational.

Proof. We first prove an elementary divisibility fact.

Claim 1. If $n \in \mathbb{Z}$ and $7 \mid n^2$, then $7 \mid n$.

By the division algorithm, applied to the integer n and the positive integer 7, there exist $k \in \mathbb{Z}$ and

$$r \in \{0, 1, 2, 3, 4, 5, 6\}$$

such that

$$n = 7k + r.$$

Squaring gives

$$n^2 = (7k + r)^2 = 49k^2 + 14kr + r^2 = 7(7k^2 + 2kr) + r^2.$$

Now suppose $7 \mid n^2$. Then there exists $t \in \mathbb{Z}$ such that

$$n^2 = 7t.$$

Using the displayed formula for n^2, we get

$$r^2 = n^2 - 7(7k^2 + 2kr) = 7t - 7(7k^2 + 2kr) = 7\big(t - 7k^2 - 2kr\big).$$

Thus $7 \mid r^2$.

But since

$$r \in \{0, 1, 2, 3, 4, 5, 6\},$$

the possible values of r^2 are

$$0, 1, 4, 9, 16, 25, 36.$$

Among these, the only number divisible by 7 is 0. Therefore $r^2 = 0$, so $r = 0$. Hence

$$n = 7k,$$

and therefore $7 \mid n$. This proves Claim 1.

We also justify the use of a fraction in lowest terms.

Claim 2. If a nonzero rational number x is given, then there exist integers p and q with $q \neq 0$ and $\gcd(p, q) = 1$ such that

$$x = \frac{p}{q}.$$

Since x is rational, there exist integers a and b with $b \neq 0$ such that

$$x = \frac{a}{b}.$$

Because $x \neq 0$, we also have $a \neq 0$. Let

$$D = \{d \in \mathbb{Z}_{>0} : d \mid a \text{ and } d \mid b\}.$$

The set D is nonempty because $1 \in D$. Also, if $d \in D$, then $d \mid a$, so $d \leq |a|$. Thus D is a nonempty finite set of positive integers, so it has a largest element. Let that largest element be d.

Since $d \mid a$ and $d \mid b$, there exist integers p and q such that

$$a = dp \qquad \text{and} \qquad b = dq.$$

Because $b \neq 0$ and $d \neq 0$, we have $q \neq 0$. Also,

$$\frac{a}{b} = \frac{dp}{dq} = \frac{p}{q}.$$

We now show that $\gcd(p, q) = 1$. Suppose c is a positive integer dividing both p and q. Then cd divides both $dp = a$ and $dq = b$, so $cd \in D$. Since d was chosen to be the largest element of D, we have

$$cd \leq d.$$

Because $d > 0$, this implies $c \leq 1$. Since c is positive, $c = 1$. Therefore the only positive common divisor of p and q is 1, so

$$\gcd(p, q) = 1.$$

This proves Claim 2.

Now suppose, for contradiction, that $\sqrt{7}$ is rational. Since $\sqrt{7} \neq 0$, Claim 2 gives integers p and q with $q \neq 0$ and

$$\gcd(p, q) = 1$$

such that

$$\sqrt{7} = \frac{p}{q}.$$

Squaring both sides gives

$$7 = \frac{p^2}{q^2}.$$

Since $q \neq 0$, we may multiply both sides by q^2 to obtain

$$p^2 = 7q^2.$$

Therefore $7 \mid p^2$. By Claim 1, it follows that

$$7 \mid p.$$

So there exists $m \in \mathbb{Z}$ such that

$$p = 7m.$$

Substituting this into $p^2 = 7q^2$ gives

$$(7m)^2 = 7q^2,$$

so

$$49m^2 = 7q^2.$$

Dividing both sides by 7 yields

$$7m^2 = q^2.$$

Equivalently,

$$q^2 = 7m^2.$$

Thus $7 \mid q^2$. Applying Claim 1 again, we get

$$7 \mid q.$$

Hence 7 divides both p and q. This contradicts the fact that

$$\gcd(p, q) = 1.$$

Therefore our assumption that $\sqrt{7}$ is rational must be false. Hence $\sqrt{7}$ is irrational. $\square$